SMARTBRIDGE
SKILLWALLET
A SmartBridge Product

Project Executable Files

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	3 Marks

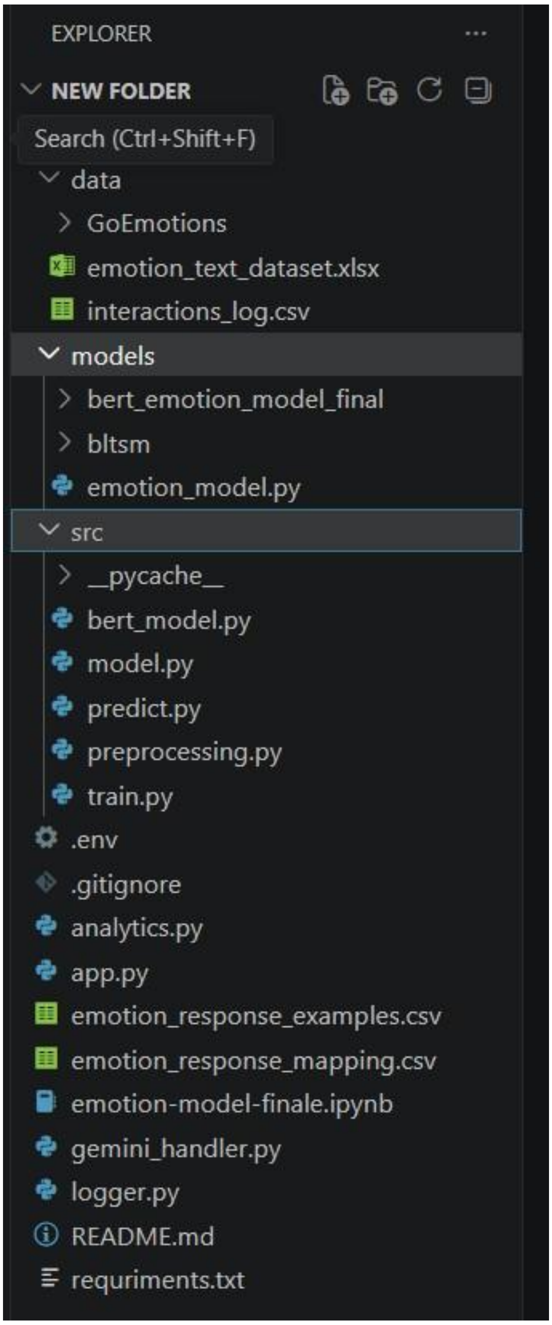
Project Executable Files

This section requires submission of all executable and deployable files for your project. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA/Partial)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	No — not yet written
3	requirements.txt / package.json / dependency file	Partial — requirement.txt exists but is missing tensorflow and torch
4	Database schema / seed files (if applicable)	NA — uses CSV file storage, no database
5	Environment configuration file (.env.example or similar)	Partial — a .env exists locally with a placeholder key, but no .env.example is committed
6	Deployed application URL (if hosted)	No — local host
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — web-based Streamlit app
8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	No — only ad-hoc __main__ test blocks exist in predictor.py and logger.py, no formal test suite
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure



Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	
Login Credentials (Demo)	Not applicable — no login is required for the current prototype
Platform / Hosting Provider	Not yet hosted — planned: Streamlit Community Cloud
Repository Link	https://github.com/yogeswaratammineni-crypto/Emotion-Detection-Learning-Support-Engine
Demo Video Link	https://drive.google.com/file/d/19SdGmrIqo09pDewHkQREtD5oumFLQ2m3/view?usp=sharing

Step 4: Run Instructions

Step 1 — Clone the repository:
git clone https://github.com/yogeswaratammineni-crypto/Emotion-Detection-LearningSupport-Engine.git cd Emotion-Detection-Learning-Support-Engine

Step 2 — Install all dependencies: pip install -r requirements.txt

Step 3 — Download model files from Google Drive and place them in exact locations:
models/bert_emotion_model_final/bert_model.pt
models/bltsm/bilstm_final.keras models/bltsm/tokenizer.pkl
models/bltsm/label_encoder.pkl

Step 4 — Configure environment variables:
Create a .env file in the root folder and add:
GEMINI_API_KEY=your_gemini_api_key_here
BILSTM_MODEL_PATH=models/bltsm/bilstm_final.keras
BERT_MODEL_PATH=models/bert_emotion_model_final/bert_model.pt
TOKENIZER_PATH=models/bltsm/tokenizer.pkl
LABEL_ENCODER_PATH=models/bltsm/label_encoder.pkl

Step 5 — Start the application: python -m streamlit run app.py

Step 6 — Open your browser at: http://localhost:8501

Step 7 — Use the application:

Type your learning challenge in the text box, select BERT or BiLSTM as the primary model, click Analyze and get guidance to detect your emotion and receive personalized Gemini AI guidance.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	A keyword-based emotion override in app.py bypasses the real BiLSTM/BERT prediction for most inputs.	Needs to be removed so the actual model output drives the response.
2	bert_model.py's bert_predict() always returns a fixed "Joy" prediction regardless of input.	Needs to be replaced with real trained BERT inference.
3	logger.py contains two incompatible logging schemas writing to the same CSV file.	Needs consolidation into a single schema.